

Rogers-Ramanujan Derivation — cycles 4-6

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Rogers-Ramanujan Derivation — cycles 4-6

Introduction

Cycles 4-6 continued milestone M2: proving the transformed-series collapse needed to complete the product-transform route for the Rogers-Ramanujan identities. Earlier cycles had already established the formal-power-series framework, the gap-two partition interpretation, the q-difference ladder, and the Euler/Jacobi product half. The open task entering cycle 4 was to prove

$$\lim_{N \rightarrow \infty} H_{\alpha, N}(q) = \sum_{j \in \mathbb{Z}} (-1)^j q^{j(5j - (2\alpha + 1))/2}, \quad \alpha \in \{0, 1\},$$

where

$$H_{\alpha, N}(q) = \sum_{n=0}^N q^{n^2 + \alpha n} \frac{(q; q)_N}{(q; q)_n}.$$

The result of cycles 4-6 is a sharper negative map of the remaining bottleneck. The signed-object model was proved, but three mechanism families were rejected: local absorb/release cancellation, pure nonlocal subset transfer, and direct fixed diagonal-window modulo-5 closure. M2 remains open.

Definitions and Notation

The finite transformed series is

$$H_{\alpha, N}(q) = \sum_{n=0}^N q^{n^2 + \alpha n} \frac{(q; q)_N}{(q; q)_n}, \quad \alpha = 0, 1.$$

Cycle 4 expanded the quotient

$$\frac{(q; q)_N}{(q; q)_n} = \prod_{s=n+1}^N (1 - q^s)$$

as a signed subset sum. Thus each term of $H_{\alpha, N}$ is represented by a signed object (n, S, α, N) , with

$$S \subseteq \{n+1, \dots, N\}, \quad w_\alpha(n, S) = n^2 + \alpha n + \sum_{s \in S} s, \quad \text{sgn}(n, S) = (-1)^{|S|}.$$

This is lemma L17 in docs/lemmas/lemma_catalogue.md, proved in docs/proof/transformed_series_collapse.md.

Cycle 4: Signed Objects and Local Cancellation

Cycle 4 targeted the remaining transformed-series collapse directly. The researcher brief (c5da6827-f96c-490e-9d57-05d2e455cf) asked for a signed-object expansion of $H_{\alpha, N}$, a candidate sign-reversing involution, and a recurrence fallback.

The worker (4f6ea7e9-9a3e-4e8f-9ec2-876d392a2e79) built:

- docs/proof/transformed_series_collapse.md
- scripts/rr/transformed_cancellation_probe.wls
- scripts/rr/plot_transformed_cancellation_survivors.py
- CSV outputs under data/finite_experiments/
- data/finite_experiments/transformed_cancellation_survivors.png

The local rule tested boundary moves that either absorb tail parts into the staircase or release parts from the staircase boundary into the tail. Weight preservation forced the equations

$$r(2n + \alpha + r)$$

for increasing n , and

$$r(2n + \alpha - r)$$

for decreasing n . Accepted moves were weight-preserving, sign-reversing, and involutive, but the rule was not total. At $N = 12$, it left 1,887 of 4,059 tested $\alpha = 0$ objects unpaired and all 3,865 tested $\alpha = 1$ objects unpaired.

The auditor (4224d822-c5e9-4052-b07b-2b85d891f2e2) validated the result as a pivot. L17 was accepted as proved, L18 was rejected as non-total, L19 remained conjectural, and L20 was accepted only as scalar stabilization.

Cycle 5: Fan-Out to Two Mechanisms

Cycle 5 pivoted away from the failed local rule. The researcher (63b892ac-9174-41ea-91c4-9ce66bee97b8) split the work into two independent branches:

- Branch A: a nonlocal sign-reversing involution on signed objects (n, S) .
- Branch B: a finite modulo-5 or shifted-state recurrence derived from $H_{\alpha, N}$ alone.

The branch objective was not another coefficient sweep. Each branch had to produce either a mechanism or a precise obstruction.

Cycle 6: Integrated Fan-Out Results

Cycle 6 integrated both fan-out outputs into the main workspace (7bf605a8-a233-4221-b34e-8d1de679900d). The integration updated:

- docs/lemmas/lemma_catalogue.md
- docs/validation.md
- promise_ledger.jsonl

Signed survivor pattern for $H_{\alpha,N}$ at $N=12$

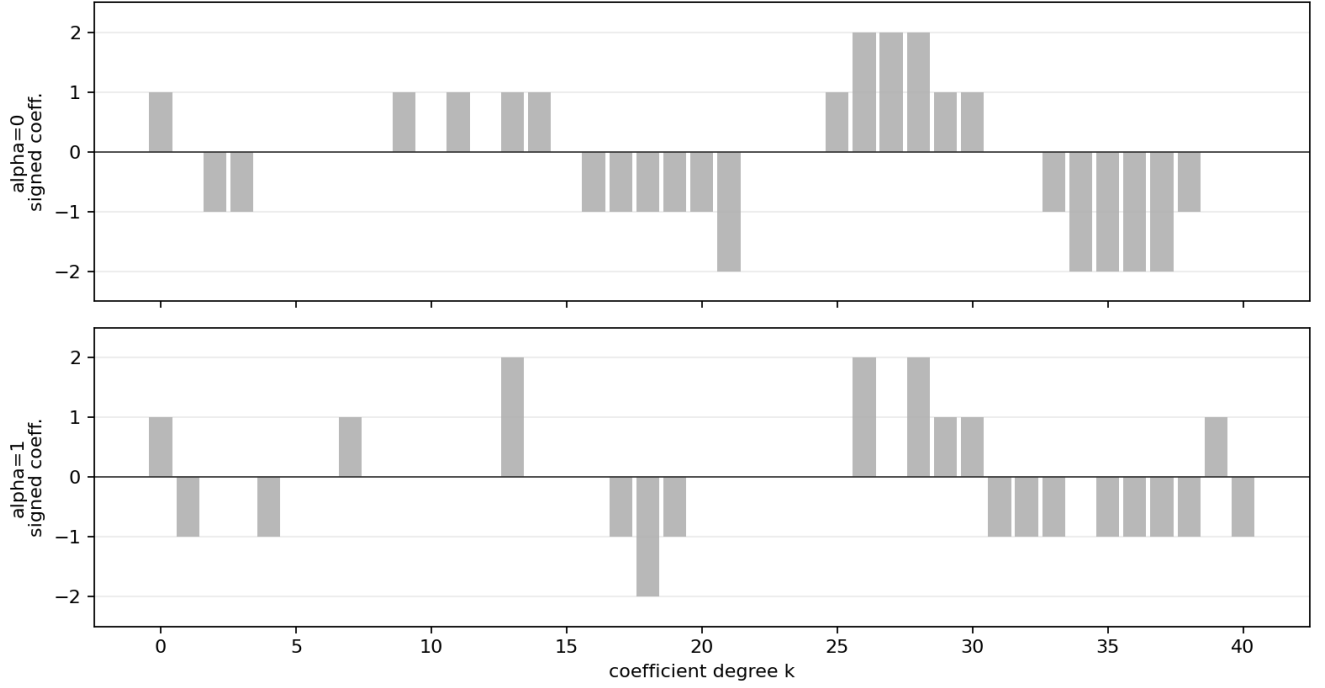


Figure 1: Signed survivor pattern for $H_{\alpha,N}$. Blue bars mark predicted pentagonal-support degrees; gray bars are finite boundary corrections.

It retained and linked:

- docs/proof/nonlocal_transformed_involution.md
- docs/proof/mod5_state_recurrence.md

The post-merge audit input validated the integration as complete, but not as proof closure.

Nonlocal Involution Branch

The nonlocal branch tested a broader pure subset-transfer rule. An upward move $n \mapsto n + r$ removes an odd subset $A \subseteq S$ satisfying

$$\sum A = r(2n + \alpha + r).$$

A downward move $n \mapsto n - r$ adds an odd missing subset satisfying

$$\sum A = r(2n + \alpha - r).$$

The probe tested deterministic smallest- r variants with lexicographic tie-breakers. Accepted transfers had no weight, sign, or tail-condition failures, but the family was rejected for two structural reasons.

First, singleton-tail objects create stable non-pentagonal no-move cases. For $\alpha = 0$, $(0, \{t\})$ can move only if $t = r^2$, so $(0, \{5\})$ is a non-pentagonal no-move object for every $N \geq 5$. For $\alpha = 1$, $(0, \{t\})$ can move only if $t = r(r + 1)$, so $(0, \{3\})$ is a non-pentagonal no-move object for every $N \geq 3$.

Second, the canonical smallest- r rule is not involutive. For example, at $\alpha = 0$, $(0, \{4\})$ moves to $(2, \emptyset)$, but the smallest reverse move from $(2, \emptyset)$ lands at $(1, \{3\})$, not the original object.

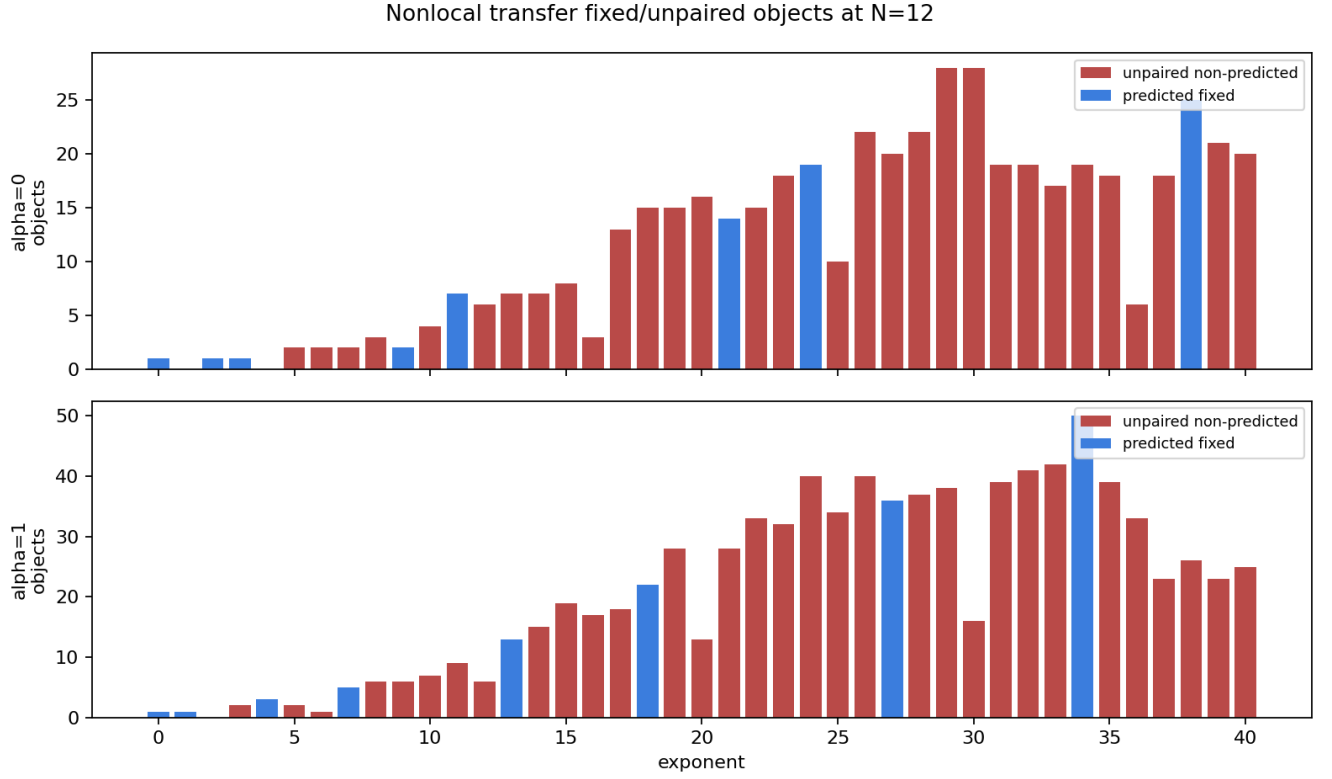


Figure 2: Fixed and unpaired objects from the nonlocal transfer search, compared against the predicted pentagonal families.

These results became L21-L23: pure nonlocal subset transfer rejected, singleton-tail obstruction proved, and canonical smallest- r transfer rejected.

Modulo-5 / Shifted-State Branch

The recurrence branch started from the scalar identity

$$H_{\alpha,N} = (1 - q^N)H_{\alpha,N-1} + q^{N^2+\alpha N}.$$

Writing

$$H_{\alpha,N}(q) = \sum_{k \geq 0} h_{\alpha,N}(k)q^k,$$

it derived the coefficient recurrence

$$h_{\alpha,N}(k) = h_{\alpha,N-1}(k) - h_{\alpha,N-1}(k - N) + \mathbf{1}_{k=N^2+\alpha N}.$$

This triangular recurrence proves coefficient stabilization: for $N > k$, the coefficient $h_{\alpha,N}(k)$ no longer changes.

For diagonal states

$$T_{\alpha,N}(d) = h_{\alpha,N}(N + d),$$

the branch proved

$$T_{\alpha,N}(d) = T_{\alpha,N-1}(d+1) - h_{\alpha,N-1}(d) + \mathbf{1}_{d=N^2+(\alpha-1)N}.$$

This exact recurrence creates the obstruction. Any fixed window $0 \leq d \leq W$ requires the outside state $T_{\alpha,N-1}(W+1)$. Iterating five steps only moves the missing boundary to $W+5$, so splitting by $N \bmod 5$ does not close the system.

Stable coefficients from H recurrence by residue class mod 5

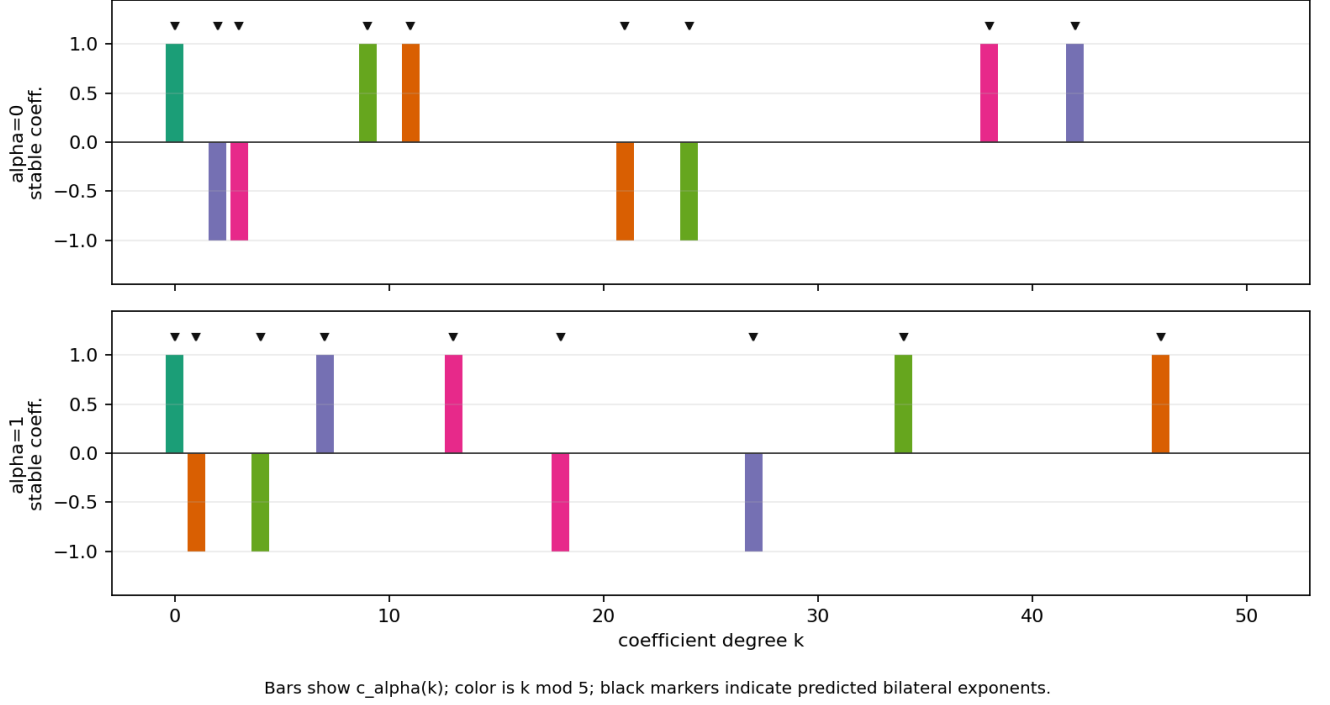


Figure 3: Stable coefficients generated from the $H_{\alpha,N}$ recurrence, colored by residue class and overlaid with predicted pentagonal exponents.

The recurrence branch produced L24-L26: coefficient recurrence proved, shifted diagonal recurrence proved, and fixed diagonal-window modulo-5 closure rejected for the direct finite-window system.

Integrated Validation

The post-merge integration reran the main and small checks for both branches.

Modulo-5 recurrence run:

```
KMax=50
NMax=60
WMax=12
Diagonal recurrence max residual: 0
Stable defects against corrected bilateral target: none_through_KMax
N=k non-stability examples: []
Finite offset window obstruction rows: 74
```

Nonlocal main probe at $N = 12$:

tie breaker	alpha	objects	paired up	paired down	predicted fixed	no move	involution failures
lex largest	0	4059	1250	1250	70	423	1066
lex smallest	0	4059	1241	1241	70	423	1084
lex largest	1	3865	714	714	131	738	1568
lex smallest	1	3865	701	701	131	738	1594

Both figure checks passed, and **promise_check** exited with warnings only. The warnings were process-scope issues: M3/M4 have not started, M2 remains active, and existing unmanaged session/report artifacts remain.

Current Proof Status

Proved during cycles 4-6:

- L17: finite signed-object expansion for $H_{\alpha,N}$.
- L20 scalar part: $H_{\alpha,N}$ stabilizes coefficientwise.
- L22: singleton-tail obstruction to pure subset-transfer cancellation.
- L24: coefficient recurrence for $H_{\alpha,N}$.
- L25: shifted diagonal recurrence.

Rejected during cycles 4-6:

- L18: local boundary absorb/release involution, because it is not total.
- L21: pure nonlocal subset-transfer rule, because stable non-pentagonal no-move objects exist.
- L23: canonical smallest- r nonlocal transfer, because it is not involutive.
- L26: direct fixed diagonal-window modulo-5 closure, because the recurrence requires outside boundary states.

Still open:

- L16/L19: the transformed-series bilateral collapse.
- M2: a complete transfer mechanism proving the Rogers-Ramanujan series equal the product/Jacobi side.

Open Questions

The next mechanism must add structure beyond the rejected families. The records specifically point to:

- a richer involution with an auxiliary marker or compensation part,
- a recurrence with additional boundary states and a real closing identity,
- or a partition-bijection route outside the transformed subset-transfer model.

Another same-axis coefficient comparison is not enough; future computation needs to propose a map, classify its failures, or prove a closed recurrence.

References

No REFERENCES.md file exists in the workspace, and the cycle records state that no web search or external proof lookup was used. No external references are cited in this report.

Appendix: Implementation Details

Session references:

cycle	researcher	worker	auditor
4	c5da6827-f96c-490e-9d57-0562e458-f91d3e-4e8f-9ec2-872249322e75	555e4f58-f91d3e-4e8f-9ec2-872249322e75	9-4052-b07b-2b85d891f

cycle	researcher	worker	auditor
5	63b892ac-9174-41ea-91c4-9ca66b1e497b	9ca66b1e497b	branch audit records integrated in cycle 6
6	post-merge integration	7bf605a8-a233-4221-b34e-8dd1de67990d	added to audit input

Code organization after integration:

path	files	role
scripts/rr/	14	Wolfram probes, Python recurrence mirror, and plotting scripts
docs/proof/	6	q-difference, finite-gap, product-transform, transformed-cancellation, nonlocal-involution, and modulo-5 recurrence notes
docs/lemmas/lemma_catalogue.md	1	Lemmas L0-L26 with proof statuses
docs/validation.md	1	Experimental checks, proof statuses, and rejected-route diagnostics
data/finite_experiments/	49	CSV outputs and figures

Key commands recorded:

```
KMAX=40 NMAX=12 OUTDIR=data/finite_experiments \
  wolfram-batch -script scripts/rr/transformed_cancellation_probe.wls
```

```
KMAX=40 NMAX=12 OUTDIR=data/finite_experiments \
  wolfram-batch -script scripts/rr/nonlocal_involution_probe.wls
```

```
KMAX=50 NMAX=60 WMAX=12 OUTDIR=data/finite_experiments \
  python3 scripts/rr/mod5_state_recurrence_probe.py
```

```
python3 -m long_exposure.tools.promise_check <run-workspace>
```

The workspace `MANIFEST.md` was refreshed for this report with the integrated script inventory, line counts, cumulative stats, and cross-references.